

Integer-Sided Triangles Meeting Three Sides of a Square

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1 Introduction

A classical problem asks for the smallest square which contains a given triangle. Jerrard and Wetzel [4] gave a complete solution. Yuan and Ding [6] also gave necessary and sufficient conditions for a triangle to fit in a square.

We use a stronger boundary condition. All three vertices of the triangle have to lie on the boundary of the square, and they have to use three different sides. More precisely, the three triangle vertices are assigned to three distinct sides of the square, one vertex to each side. A vertex at a corner may be assigned to either of the two sides which meet there.

We also require the three side lengths of the triangle to be positive integers. Two triangles are counted as the same if they are congruent.

For a positive integer n , let $a(n)$ be the number of these triangles for a square of side length n . The sequence is OEIS A400509.

The first terms are

1, 2, 6, 10, 19, 28, 42, 58, 82, 108, 143, 176, 223, 271, 331, 397,
472, 550, 644, 738, 849, 971, 1108, 1239, 1405, 1567, 1747, 1935,
2149, 2367,

For example, $a(1) = 1$. The equilateral triangle of side 1 works. One placement has vertices

$$\left(1 - \frac{\sqrt{3}}{2}, 0\right), \quad \left(1 - \frac{\sqrt{3}}{2}, 1\right), \quad \left(1, \frac{1}{2}\right).$$

For $n = 3$, the six triangles are

$$(1, 3, 3), (2, 2, 3), (2, 3, 3), (2, 3, 4), (3, 3, 3), (3, 3, 4).$$

Here and below, a triple such as $(2, 3, 4)$ gives the three side lengths in increasing order.

Our geometry is closely related to the square types in [4]. There is one important difference. In the ordinary containment problem a longest side may be a diagonal of the square. Then the third vertex can lie inside the square. This is not allowed here. In fact, this diagonal-square region is the only region where the two minimum-square problems differ.

We first solve the geometric problem for one triangle. Then we count the integer triangles. Finally we study the growth of $a(n)$.

2 An exact contact test

Let a triangle have vertices A, B, C , with $AB = q$, $AC = r$, and $BC = s$. We first ask for a placement in which A and B lie on two opposite sides of a square of side length n . The vertex C must lie on a third side.

Put

$$H = \sqrt{(q+r+s)(-q+r+s)(q-r+s)(q+r-s)}.$$

Thus $H = 4\Delta$, where Δ is the area of the triangle. Also put

$$U = q^2 + r^2 - s^2, \quad V = q^2 + s^2 - r^2.$$

If $q \geq n$, put

$$t = \frac{\sqrt{q^2 - n^2}}{n}.$$

Theorem 1. *The required placement exists, with A and B on two opposite sides and C on a third side, if and only if*

$$n \leq q \leq \sqrt{2}n$$

and

$$0 \leq U + Ht \leq 2q^2, \tag{1}$$

$$0 \leq H - Ut \leq 2q^2, \tag{2}$$

$$0 \leq H + Vt \leq 2q^2. \tag{3}$$

Proof. Put the two endpoints of the side q at

$$A = (0, y), \quad B = (n, y + d),$$

where

$$d = \sqrt{q^2 - n^2}.$$

Put the third vertex at

$$C = (x, 0).$$

The projection of AC on AB has length

$$u = \frac{q^2 + r^2 - s^2}{2q} = \frac{U}{2q}.$$

The height of the triangle over the side q is

$$h = \frac{H}{2q}.$$

Resolving the vector AC in the directions parallel and perpendicular to AB gives

$$x = \frac{nU + dH}{2q^2}$$

and

$$y = \frac{nH - dU}{2q^2}.$$

Since $t = d/n$,

$$\frac{x}{n} = \frac{U + Ht}{2q^2}, \quad \frac{y}{n} = \frac{H - Ut}{2q^2}.$$

Also $U + V = 2q^2$, so

$$\frac{y + d}{n} = \frac{H + Vt}{2q^2}.$$

Thus (1)–(3) say exactly that

$$0 \leq x \leq n, \quad 0 \leq y \leq n, \quad 0 \leq y + d \leq n.$$

These are exactly the required conditions.

Finally, the distance between two points on opposite sides of a square of side n is between n and $\sqrt{2}n$. This gives the first inequality. $\square$

For a given triangle there are only six tests: three choices for q , and two orders for r, s .

Example 1. Take the 3, 4, 5 triangle. Theorem 1 reduces the problem to six algebraic checks. No search over angles or boundary points is needed. We return to this triangle in Example 3, where we find its smallest admissible square.

3 There are no gaps

We now fix the triangle and vary the square.

Theorem 2. *Let T be a nondegenerate triangle, and let c be its longest side. There is a number $m(T)$ such that T has three-side contact with a square of side length n if and only if*

$$m(T) \leq n \leq c.$$

Proof. Scale the square to side length 1. Fix a side q of the triangle. Put

$$u = \frac{q^2 + r^2 - s^2}{2q^2}, \quad h = \frac{2\Delta}{q^2}.$$

If the scaled length of q is $\sqrt{1+t^2}$, the three vertices can be written, after a reflection if needed, as

$$A = (0, h - ut),$$

$$B = (1, h + (1 - u)t),$$

$$C = (u + ht, 0).$$

Hence this contact type is possible exactly when

$$0 \leq u + ht \leq 1, \tag{4}$$

$$0 \leq h - ut \leq 1, \tag{5}$$

$$0 \leq h + (1 - u)t \leq 1. \tag{6}$$

All six inequalities are linear in t . Thus the possible values of t for one fixed contact type form an interval. The same is true for the scale factor

$$\lambda = \frac{\sqrt{1+t^2}}{q}.$$

We show that the intervals for different sides cannot be separated.

First suppose that the left endpoint of one interval has $t = 0$. Then

$$A = (0, h), \quad B = (1, h), \quad C = (u, 0),$$

with $0 \leq u \leq 1$ and $0 < h \leq 1$. If AB is not a longest side, assume, after reflection if needed, that

$$p = AC = \sqrt{u^2 + h^2} > 1.$$

Put

$$k = \sqrt{p^2 - 1},$$

$$\alpha = \frac{u - hk}{p^2}, \quad \beta = \frac{h + uk}{p^2}.$$

There is a rotation which sends the vector $(u, -h)$ to $(1, k)$. Its cosine and sine are α and β . Hence it sends the vector $(1, 0)$ to (α, β) .

It remains only to check that the rotated triangle lies in the unit square. Since $u, h \leq 1$, we have $k \leq u$ and $k \leq h$. Hence $\alpha \geq 0$, and clearly $\alpha \leq 1$. Also

$$(p^2 - h)^2 - u^2 k^2 = p^2 (h - 1)^2 \geq 0.$$

Since $p^2 - h > 0$, this gives $\beta \leq 1$. Finally,

$$p^2 - u = h^2 + u^2 - u \leq h^2 \leq h,$$

and $k \leq 1$. Thus $k(p^2 - u) \leq h$, which is the same as $\beta \geq k$. After a translation, the rotated triangle therefore has vertices

$$A' = (0, 1 - \beta), \quad C' = (1, 1 - \beta + k), \quad B' = (\alpha, 1).$$

All three points lie in the unit square. The side $A'C'$ has length $p > 1$ and joins the two opposite vertical sides. Thus the left endpoint also belongs to an interval for a strictly longer triangle side.

Now suppose that the left endpoint has $t > 0$. A nonempty interval in (4)–(6) must have $u \leq 1$ and $h \leq 1$. If $u \geq 0$, then $t = 0$ also satisfies (4)–(6). Hence $u < 0$. The left endpoint is therefore

$$t = -\frac{u}{h},$$

where $u + ht = 0$. At this endpoint write

$$C = (0, 0), \quad A = (0, Y), \quad B = (1, Z).$$

Then

$$Y = h + \frac{u^2}{h}, \quad Z = Y - \frac{u}{h},$$

and

$$0 < Y < Z \leq 1.$$

The present opposite-side length is

$$AB = \sqrt{1 + (Z - Y)^2}.$$

But

$$BC = \sqrt{1 + Z^2} > AB.$$

We now use BC as the opposite-side length. Rotate the vector $(1, Z)$ to $(1, -Z)$. Under this rotation the vector $(0, Y)$ becomes (ξ, η) , where

$$\xi = \frac{2YZ}{1 + Z^2}, \quad \eta = \frac{Y(1 - Z^2)}{1 + Z^2}.$$

Since $0 < Y < Z \leq 1$,

$$0 \leq \xi \leq 1$$

and

$$\eta + Z \leq \frac{Z(1 - Z^2)}{1 + Z^2} + Z = \frac{2Z}{1 + Z^2} \leq 1.$$

Thus

$$C' = (0, 1 - \eta), \quad B' = (1, 1 - \eta - Z), \quad A' = (\xi, 1)$$

is again a placement in the same unit square, now with the strictly longer side BC joining the opposite sides.

We can therefore move from the left endpoint of any contact interval to an overlapping interval belonging to a strictly longer triangle side. There are only three side lengths. After at most two such steps we reach a longest side.

For a longest side c , the value $t = 0$ is possible. Indeed, for $q = c$ we have $0 \leq u \leq 1$, because c is at least as long as each of the other two sides. Also the altitude to c is at most either adjacent side, and hence at most c . Thus $h \leq 1$. Therefore (4)–(6) hold at $t = 0$.

The union of all contact intervals in the scale variable λ is therefore one interval starting at $1/c$.

Returning to the original square size $n = 1/\lambda$, the possible values form one interval ending at c . We call its lower endpoint $m(T)$. $\square$

Thus every fixed triangle has a smallest admissible square. Its largest admissible square has side c , and every intermediate square size also works.

Example 2. For an equilateral triangle of side k ,

$$m(T) = k \cos 15^\circ = \frac{\sqrt{6} + \sqrt{2}}{4} k.$$

Thus the triangle has three-side contact with a square of side n exactly when

$$k \cos 15^\circ \leq n \leq k.$$

For $k = 1$, this is the first example from the introduction.

Example 3. For a right triangle with legs $p \leq q$,

$$m(T) = \frac{q^2}{\sqrt{q^2 + (q - p)^2}}.$$

For the 3, 4, 5 triangle this is

$$m(T) = \frac{16}{\sqrt{17}} = 3.880570 \dots$$

Thus the possible square side lengths are

$$\left[\frac{16}{\sqrt{17}}, 5 \right].$$

4 The minimum square for a normalized triangle

Theorem 2 leaves one question: how small can the square be? Since the answer scales with the triangle, we normalize the longest side to 1. Write the side lengths as

$$(x, y, 1),$$

where

$$0 < x \leq y \leq 1, \quad x + y > 1.$$

Let $\rho(x, y)$ be the largest factor by which this triangle can be enlarged and still have three-side contact with the unit square. Then

$$m(x, y, 1) = \frac{1}{\rho(x, y)}.$$

Put

$$W = \sqrt{4x^2y^2 - (x^2 + y^2 - 1)^2},$$

$$U = 1 + x^2 - y^2, \quad V = 1 + y^2 - x^2, \quad S = x^2 + y^2 - 1.$$

The identities

$$U^2 + W^2 = 4x^2, \quad V^2 + W^2 = 4y^2, \quad S^2 + W^2 = 4x^2y^2 \quad (7)$$

will be useful.

Different shapes have different optimal contacts with the square. The shape region splits into four parts. Figure 1 shows these parts; the four lines in the next theorem correspond to Regions I–IV.

Theorem 3. For $0 < x \leq y \leq 1$ and $x + y > 1$,

$$\rho(x, y) = \begin{cases} \frac{2x}{U}, & W \leq U, \\ \frac{2\sqrt{1+y^2-W}}{V}, & U \leq W \leq V, \\ \frac{2x}{W}, & V \leq W \text{ and } W \geq 2x^2, \\ \frac{2\sqrt{x^2+y^2-W}}{S}, & V \leq W \leq 2x^2. \end{cases} \quad (8)$$

At the common boundaries the neighboring formulas have the same value. At the right isosceles point the formula is read by continuity.

Proof. Let $A \leq B \leq C$ be the angles opposite $x, y, 1$. Then

$$\tan B = \frac{W}{U}, \quad \tan A = \frac{W}{V}.$$

Thus

$$W \leq U \iff B \leq 45^\circ,$$

and

$$W \leq V \iff A \leq 45^\circ.$$

We first take the region $B \leq 45^\circ$. Then $C \geq 90^\circ$, so

$$S = x^2 + y^2 - 1 \leq 0.$$

Also $W \leq U \leq V$ and $U + V = 2$.

Take first $q = 1$ and the order $r = x, s = y$. Then $u = U/2$ and $h = W/2$. The three upper bounds for t in (4)–(6) are

$$\frac{V}{W}, \quad \frac{W}{U}, \quad \frac{2 - W}{V}. \quad (9)$$

The middle one is the smallest. Indeed,

$$W^2 \leq UV$$

because $W \leq U \leq V$, and

$$WV \leq U(2 - W)$$

is equivalent, using $U + V = 2$, to $W \leq U$. Thus

$$t_{\max} = \frac{W}{U}.$$

If r and s are interchanged, the active bound is W/V , which is no larger.

It remains to exclude a better contact with $q = x$ or $q = y$. For $q = x$ and the order $r = y, s = 1$, we have

$$u = \frac{S}{2x^2} \leq 0, \quad h = \frac{W}{2x^2}.$$

Conditions (4) and (6) require

$$t \geq -\frac{u}{h}, \quad t \leq \frac{1 - h}{1 - u}.$$

For these bounds to be compatible we need

$$u^2 + h^2 - u - h \leq 0.$$

But a direct calculation gives

$$u^2 + h^2 - u - h = \frac{V - W}{2x^2} \geq 0.$$

Thus this contact is impossible in the interior of the region. At the boundary it can only tie. If r and s are interchanged, u is replaced by $1 - u \geq 1$, and (4) is impossible for $t > 0$.

The same calculation for $q = y$ gives the necessary condition $U \leq W$. Hence this contact is impossible when $W < U$. On the boundary $W = U$, the first formula below gives $\rho = \sqrt{2}$. The competing contact with $q = y$ has the same scale and does not improve it. For $q = x$, equality in the compatibility condition would require $V = W$. Together with $W \leq U \leq V$, this is possible only at the right isosceles point. Thus no shorter side gives a larger scale. We obtain

$$\rho = \sqrt{1 + (W/U)^2} = \frac{2x}{U},$$

where the last equality follows from (7). This is the first line of (8).

For the other three regions we use the classification of smallest containing squares in [4, Theorem 8]. Their diagonal-square case is exactly the region $W \leq U$, already treated above. In the other three regions the minimum is a vertex square or a side square. A vertex square already has our three-side contact. In the side-square region the triangle is acute or right, so the perpendicular from the opposite vertex meets the shortest side

segment. We can slide the triangle parallel to this side until one endpoint reaches a corner. The other two vertices stay on their square sides. We then have three-side contact. Hence in these three regions our minimum is the same as in [4].

Suppose first that

$$A \leq 45^\circ \leq B,$$

or equivalently $U \leq W \leq V$. The smallest square is the vertex square at the vertex of angle A . The vertex-square formula of [4] gives

$$m = \frac{V}{2\sqrt{1+y^2-W}}.$$

Therefore

$$\rho = \frac{2\sqrt{1+y^2-W}}{V}.$$

This is the second line of (8).

Now suppose that $A \geq 45^\circ$, so $W \geq V$. The triangle is tall on its shortest side x exactly when the height to x is at least x . Since this height is $W/(2x)$, the condition is

$$W \geq 2x^2.$$

In this case the smallest square is the side square on x , with side

$$m = \frac{W}{2x}.$$

Thus

$$\rho = \frac{2x}{W},$$

which is the third line.

Finally suppose that $W \geq V$ and $W \leq 2x^2$. The triangle is not tall. The smallest square is then the vertex square at the largest angle C . The vertex-square formula gives

$$m = \frac{S}{2\sqrt{x^2+y^2-W}}.$$

Hence

$$\rho = \frac{2\sqrt{x^2+y^2-W}}{S}.$$

This is the fourth line.

The formulas agree when $B = 45^\circ$, when $A = 45^\circ$, and when the height to x is equal to x . Thus the four cases fit together. $\square$

Let $A \leq B \leq C$ be the angles opposite $x, y, 1$. Since

$$\tan B = \frac{W}{U}, \quad \tan A = \frac{W}{V},$$

the first two boundaries have a simple meaning:

$$W = U \iff B = 45^\circ,$$

$$W = V \iff A = 45^\circ.$$

The third boundary

$$W = 2x^2$$

means that the height to the shortest side x is equal to x .

This four-part division is close to the four-region division in [4]. In their ordinary containment problem, the diagonal square is optimal in one region. In our problem that square is not allowed, because the third triangle vertex need not be on the square boundary. This is why the first formula in (8) is different.

Figure 1 shows the four regions. The outer triangle is the shape region

$$D = \{(x, y) : 1/2 < y \leq 1, 1 - y < x \leq y\}.$$

The three internal curves can also be written as

$$W = U \iff y^2 = 1 + x^2 - \sqrt{2}x,$$

$$W = V \iff x^2 = 1 + y^2 - \sqrt{2}y,$$

and, on the relevant branch,

$$W = 2x^2 \iff y^2 = 1 + x^2 - 2x\sqrt{1 - x^2}.$$

They meet at the right isosceles point

$$P = (2^{-1/2}, 2^{-1/2}).$$

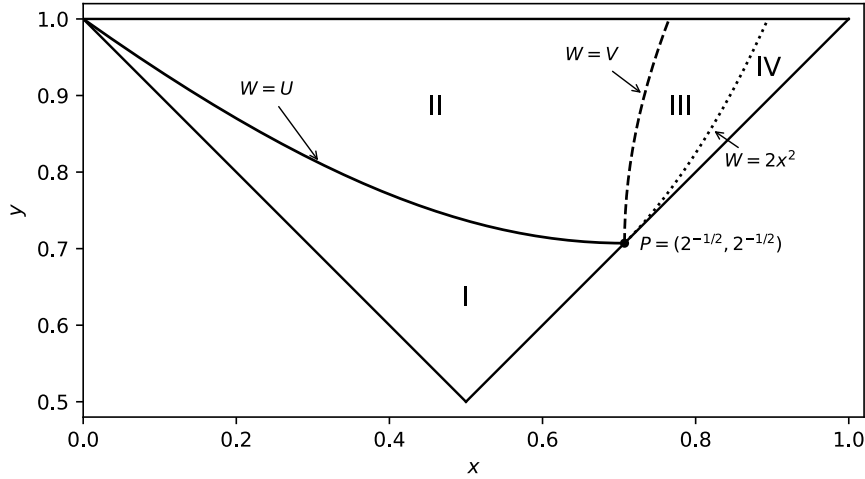


Figure 1: The four regions in Theorem 3. Regions I–IV correspond to the four lines of (8). Region I is the diagonal-square region of the ordinary containment problem; it is the only region where that problem and the present three-side contact problem have different minima.

5 The integer sequence

Every congruence class of an integer-sided triangle has one unique ordered triple

$$1 \leq a \leq b \leq c,$$

with

$$a + b > c.$$

By Theorem 2, it is counted by $a(n)$ exactly when

$$m(a, b, c) \leq n \leq c.$$

Also, two triangle vertices lie on opposite sides of the square. Hence

$$c \leq \sqrt{2}n.$$

Corollary 4. *The number $a(n)$ is the number of integer triples*

$$1 \leq a \leq b \leq c, \quad a + b > c,$$

for which

$$m(a, b, c) \leq n \leq c.$$

In particular,

$$n \leq c \leq \sqrt{2}n.$$

This gives a finite algorithm. One may use either Theorem 1 or Theorem 3. For numerical work there is one small problem. Near the right isosceles point, the fourth line of (8) subtracts nearly equal numbers. We therefore use the equivalent form

$$\rho(x, y)^2 = \frac{4((x^2 - y^2)^2 + S^2)}{S^2(x^2 + y^2 + W)},$$

with the value $\rho = \sqrt{2}$ at $S = 0$.

Some values are

$$a(10) = 108, \quad a(20) = 738, \quad a(50) = 10448, \quad a(100) = 80606.$$

For these computations we used Theorem 1. Boundary equalities were tested exactly; we squared an inequality only after its signs were known. As a check, we also computed all values through $n = 100$ from Theorem 3. The two computations agree. A verification program is included with the source files.

Example 4. The longest side does not always join the two opposite square sides. The triangle

$$(34, 34, 35)$$

is counted for $n = 33$, but a side of length 34 must join the opposite sides.

6 The main asymptotic term

The formulas above turn the original problem into a lattice-point count. Let R be the set of real triples

$$0 < x \leq y \leq z, \quad x + y > z,$$

which have three-side contact with the unit square. Scaling gives

$$a(n) = \#(nR \cap \mathbb{Z}^3). \tag{10}$$

The set R is bounded, with

$$1 \leq z \leq \sqrt{2}.$$

It is also semialgebraic by Theorem 1.

Davenport's Lipschitz principle [3] now gives the first asymptotic formula.

Theorem 5. *There is a positive constant C such that*

$$a(n) = Cn^3 + O(n^2).$$

Moreover,

$$C = \text{vol}(R).$$

The constant has a simple two-dimensional integral. Normalize the longest side to 1. The shape region is

$$D = \{(x, y) : 1/2 < y \leq 1, 1 - y < x \leq y\}.$$

For a fixed shape, Theorem 2 gives

$$1 \leq z \leq \rho(x, y).$$

The change of variables

$$(a, b, c) = (zx, zy, z)$$

has Jacobian z^2 . Hence

$$C = \frac{1}{3} \int_{1/2}^1 \int_{1-y}^y (\rho(x, y)^3 - 1) \, dx \, dy. \quad (11)$$

Using (8), numerical integration gives

$$C = 0.0776924549 \dots$$

7 The second asymptotic term

We now keep one more term. This is the only part of the paper which needs more than the elementary geometry above. We cut the three-dimensional region into horizontal planar sections. In each section, the curved boundary gives a smaller error. The two rational straight edges give the next term after the sections are summed.

For $1 \leq z \leq \sqrt{2}$, let

$$D_z = \{(x, y) : (x, y, z) \in R\},$$

and let

$$F(z) = \text{area}(D_z).$$

We first record the shape of these sections.

Lemma 6 (Boundary of a section). *For $1 < z < \sqrt{2}$, the boundary of D_z consists of finitely many algebraic arcs and at most two straight segments. The straight segments lie on*

$$x = y \quad \text{or} \quad y = z.$$

There is no straight boundary segment on $x + y = z$, and no level arc coming from $\rho = z$ is a line segment.

Proof. Write

$$(x, y, z) = z(X, Y, 1).$$

Then

$$0 < X \leq Y \leq 1, \quad X + Y > 1,$$

and Theorems 2 and 3 say that

$$1 \leq z \leq \rho(X, Y).$$

Thus the non-linear part of the boundary comes from

$$\rho(X, Y) = z.$$

On each of the four regions in Theorem 3, this equation is algebraic after the square roots are removed. The degrees are bounded by an absolute constant. Hence each section has only finitely many algebraic boundary arcs.

The level arcs are not line segments. This can be checked from the four square types. In Region I the equation in the unscaled variables is

$$(x - 1)^2 - y^2 = 1 - z^2,$$

a nondegenerate hyperbola for $z > 1$. In Region III the minimum is a side square. Put one endpoint of the shortest side at $(0, 0)$ and the opposite vertex at $(d, 1)$, where

$$d = \sqrt{z^2 - 1}.$$

If the shortest side has length x , the remaining side on this boundary satisfies

$$y^2 = 1 + (x - d)^2,$$

up to reflection. This is again not a line.

In Region II the corner vertex can be put at $(0, 0)$. Put the endpoint of the side of length z at $(1, d)$, and the other vertex at $(u, 1)$. Then

$$x(u)^2 = (1 - u)^2 + (1 - d)^2, \quad y(u)^2 = 1 + u^2.$$

Neither coordinate is constant on an open interval. Moreover,

$$\left(\frac{x'(u)}{y'(u)} \right)^2 = \frac{(1 - u)^2(1 + u^2)}{u^2((1 - u)^2 + (1 - d)^2)}.$$

This cannot be constant on an interval: after cross multiplication, the left side of the resulting polynomial identity has a nonzero constant term, while the right side has a factor u^2 . Hence no open part of this curve is a line segment.

In Region IV put the corner vertex at $(0, 0)$ and the other two vertices at $(u, 1)$ and $(1, v)$. Then

$$(1 - u)^2 + (1 - v)^2 = z^2,$$

and the other two side lengths are $\sqrt{1 + u^2}$ and $\sqrt{1 + v^2}$. We can also use the polynomial level equation in the proof of Lemma 7 below. Its highest homogeneous part in the two shape variables is

$$z^4(X^2 + Y^2)^4.$$

This does not vanish identically on any real line. Hence the Region IV level curve has no line segment.

The order conditions $X \leq Y \leq 1$ give the two possible straight parts $x = y$ and $y = z$.

It remains to consider the triangle inequality. On the degenerate boundary $X + Y = 1$, we have $W = 0$ and $U = 2X$. The first line of (8), read at the boundary, gives $\rho = 2X/U = 1$. Hence $z \leq \rho(X, Y)$ cannot hold near this boundary when $z > 1$. Therefore $x + y = z$ is not a boundary segment of D_z for $1 < z < \sqrt{2}$. $\square$

For a segment in a rational direction, we use its lattice-normalized length. Thus a segment of Euclidean length $\ell\sqrt{2}$ on $x = y$ has normalized length ℓ , while a horizontal segment has its usual length. Let $L(z)$ be the total normalized length of the straight parts of ∂D_z .

Lemma 7 (Uniform slice estimate). *There is a finite set $E \subset (1, \sqrt{2})$ with the following property. If J is a closed interval contained in $(1, \sqrt{2}) \setminus E$, then there is a number $\delta > 0$ such that, uniformly for $k/n \in J$,*

$$\#(nD_{k/n} \cap \mathbb{Z}^2) = n^2 F(k/n) + \frac{n}{2} L(k/n) + O(n^{1-\delta}). \quad (12)$$

Proof. We need the estimate uniformly in z , so we give the details.

The total boundary, with z also treated as a variable, is a fixed semialgebraic set. First we make the algebraic complexity explicit. Put

$$\mathcal{W} = 4X^2Y^2 - (X^2 + Y^2 - 1)^2.$$

Before the region inequalities are imposed, the four level equations $\rho(X, Y) = z$ are contained in the zero sets of

$$\begin{aligned} & z(1 + X^2 - Y^2) - 2X, \\ & (4(1 + Y^2) - z^2(1 + Y^2 - X^2)^2)^2 - 16\mathcal{W}, \\ & z^2\mathcal{W} - 4X^2, \end{aligned}$$

and

$$(4(X^2 + Y^2) - z^2(X^2 + Y^2 - 1)^2)^2 - 16\mathcal{W}.$$

Thus the degrees are bounded by an absolute constant. A finite semialgebraic stratification gives a finite partition of the parameter interval $(1, \sqrt{2})$ into open intervals and points. After the finitely many point parameters are put into E , the curved boundary over each remaining interval is a fixed finite collection of regular real analytic algebraic arcs which vary analytically with z . Their number and their degrees are bounded independently of z .

Next we check that the finite type is uniform. Let one regular irreducible arc be given by

$$P(x, y) = 0,$$

where P has degree d . The numerator of its curvature is

$$K_P = P_{yy}P_x^2 - 2P_{xy}P_xP_y + P_{xx}P_y^2.$$

It has degree at most $3d - 4$. By Lemma 6, the arc is not a line, so P does not divide K_P . Bézout's theorem therefore bounds the intersection multiplicity of $P = 0$ and $K_P = 0$ in terms of d only. In particular, the order of every zero of the curvature is bounded by one integer which is valid for the whole family.

For a planar analytic boundary whose curvature has zeros of bounded finite order, the classical finite-type lattice-point estimates give a power saving over the trivial $O(n)$ boundary error. Colin de Verdière [2, Theorem 1] gives

$$O(n^{1-1/(m+2)})$$

when m is an upper bound for the order of a curvature zero; in the nonvanishing-curvature case the usual $O(n^{2/3})$ estimate is enough. Müller and Nowak [5, pp. 173–174] use boundary weight $1/2$ and summarize the same finite-type estimate.

We apply this estimate to our sections. Cut D_z at its finitely many corners and vertical tangencies. Each resulting part has upper and lower boundaries which are analytic graphs. With boundary lattice points given weight $1/2$, counting one vertical lattice column gives the area term plus the usual sawtooth term for each graph. Internal cuts occur twice and cancel. The curved graph pieces have uniformly bounded finite type by the preceding paragraph, so their sawtooth sums are $O(n^{1-\delta})$ for some $\delta > 0$. The pieces form a compact family when $z \in J$, and the constants can therefore be chosen uniformly. The finitely many cut points change the count by only $O(1)$.

Let $N^*(n, z)$ denote this half-weighted lattice count. The two straight graph pieces from Lemma 6 give no linear discrepancy in the half-weight convention. Indeed, when $z = k/n$, after dilation they lie on $x = y$ and $y = k$. Their graph values at integer columns are integers, so the corresponding half-weight sawtooth terms vanish. Thus

$$N^*(n, z) = n^2 F(z) + O(n^{1-\delta})$$

for some $\delta > 0$, uniformly for $z \in J$.

Now return to full boundary weight. The two straight segments contain

$$nL(z) + O(1)$$

lattice points. The curved algebraic arcs contain $o(n)$ lattice points uniformly. To see this, clear the denominators in the fixed level equations above and factor them into components. A degree-one real factor would give a line segment and is excluded by Lemma 6. Every remaining real component has degree at least 2, bounded independently of z . For an absolutely irreducible component, Bombieri and Pila [1, Theorem 5] give a sublinear bound depending only on this degree. If a component defined over $\mathbb{Q}$ is not absolutely irreducible, its rational points lie in intersections of conjugate components, so Bézout gives only $O(1)$ such points. Thus the same sublinear bound is enough for the whole curved boundary. Hence changing the boundary weight from $1/2$ to 1 adds

$$\frac{n}{2}L(z) + O(n^{1-\delta'})$$

for some $\delta' > 0$. Replacing δ by the smaller exponent proves (12). $\square$

Remark 8. The half weight in the proof is important. A rational straight segment can contain a number of lattice points proportional to n . A curved algebraic arc of fixed degree contains only $o(n)$ lattice points. This is why only the straight parts appear in the second term below.

Theorem 9. *We have*

$$a(n) = Cn^3 + Bn^2 + o(n^2),$$

where C is given by (11) and

$$B = \frac{1}{8} + \frac{1}{4} \int_{1/2}^1 (\rho(t, t)^2 - 1) dt + \frac{1}{4} \int_0^1 (\rho(t, 1)^2 - 1) dt. \quad (13)$$

Numerically,

$$B = 0.2878839246 \dots$$

Proof. From (10), slice by the last coordinate. We get

$$a(n) = \sum_{k=n}^{\lfloor \sqrt{2}n \rfloor} \#(nD_{k/n} \cap \mathbb{Z}^2). \quad (14)$$

Fix $\varepsilon > 0$. Take small open intervals around the finite set E in Lemma 7 and also around the two endpoints 1 and $\sqrt{2}$. Choose them so that their total length is at most ε . The remaining parameter set is a finite union of closed intervals J contained in $(1, \sqrt{2}) \setminus E$. On each of them, (12) is uniform. Summing its error over $O(n)$ sections gives $o(n^2)$.

For the sections in the removed intervals we use the ordinary planar Lipschitz estimate. The family D_z has fixed semialgebraic complexity and bounded diameter. Hence its boundary length, and the constant in the $O(n)$ Lipschitz error, are bounded uniformly in z . Also $L(z) = O(1)$ uniformly. There are at most $\varepsilon n + O(1)$ such sections. Their total error is therefore

$$O(\varepsilon n^2) + O(n).$$

First let $n \rightarrow \infty$, and then let $\varepsilon \rightarrow 0$. This proves that the total error in (14), after the area and straight-boundary terms are removed, is $o(n^2)$.

It remains to sum these two main terms.

The function F is continuous and semialgebraic. Hence it is piecewise C^1 and its derivative is integrable. Also

$$F(1) = \frac{1}{4}, \quad F(\sqrt{2}) = 0.$$

Indeed, when $z = 1$, every triangle with longest side 1 works, so

$$D_1 = \{(x, y) : 0 < x \leq y \leq 1, x + y > 1\},$$

which has area $1/4$. At $z = \sqrt{2}$ the section may contain boundary points, but it has area zero. The first Euler summation formula, with $F(\sqrt{2}) = 0$, therefore gives

$$\sum_{k=n}^{\lfloor \sqrt{2}n \rfloor} F(k/n) = n \int_1^{\sqrt{2}} F(z) dz + \frac{1}{8} + o(1). \quad (15)$$

The integral is the volume C .

Now consider the two straight parts. The line $x = y$ gives the flat surface

$$(x, y, z) = (zt, zt, z), \quad \frac{1}{2} \leq t \leq 1, \quad 1 \leq z \leq \rho(t, t).$$

The Euclidean area element is $\sqrt{2}z dt dz$. The lattice in the plane $x = y$ has covolume $\sqrt{2}$, so the lattice-normalized area element is $z dt dz$. Hence its total normalized area is

$$K_1 = \frac{1}{2} \int_{1/2}^1 (\rho(t, t)^2 - 1) dt. \quad (16)$$

Similarly, the plane $y = z$ has the same lattice covolume $\sqrt{2}$, and we get

$$K_2 = \frac{1}{2} \int_0^1 (\rho(t, 1)^2 - 1) dt. \quad (17)$$

Therefore

$$\int_1^{\sqrt{2}} L(z) dz = K_1 + K_2.$$

The function L is bounded and piecewise continuous. A Riemann sum gives

$$\frac{n}{2} \sum_{k=n}^{\lfloor \sqrt{2}n \rfloor} L(k/n) = \frac{1}{2}(K_1 + K_2)n^2 + o(n^2). \quad (18)$$

Combining (15)–(18) gives

$$B = \frac{1}{8} + \frac{K_1 + K_2}{2},$$

which is (13). □

The two one-dimensional functions in (13) are explicit. We have

$$\rho(t, t) = \begin{cases} 2t, & \frac{1}{2} \leq t \leq \frac{1}{\sqrt{2}}, \\ \frac{2\sqrt{2}}{1 + \sqrt{4t^2 - 1}}, & \frac{1}{\sqrt{2}} \leq t \leq 1, \end{cases}$$

so

$$K_1 = 0.1145847027810355 \dots$$

Also

$$\rho(t, 1) = \begin{cases} \frac{2\sqrt{2 - t\sqrt{4 - t^2}}}{2 - t^2}, & 0 \leq t \leq \sqrt{2 - \sqrt{2}}, \\ \frac{2}{\sqrt{4 - t^2}}, & \sqrt{2 - \sqrt{2}} \leq t \leq \frac{2}{\sqrt{5}}, \\ \frac{2\sqrt{1 + t^2 - t\sqrt{4 - t^2}}}{t^2}, & \frac{2}{\sqrt{5}} \leq t \leq 1, \end{cases}$$

which gives

$$K_2 = 0.2111831463706464 \dots$$

Thus

$$B = \frac{1}{8} + \frac{K_1 + K_2}{2} = 0.287883924575841 \dots$$

8 Final remarks

The extra boundary condition changes the problem in a real way. This can already be seen for the triangle $(2, 2, 3)$. Its longest side can be a diagonal of a square of side $3/\sqrt{2}$, but then the third vertex is inside the square. For three-side contact, Theorem 3 gives

$$m(2, 2, 3) = \frac{9}{4}.$$

Theorem 2 is useful beyond the integer problem. It says that every fixed triangle has one interval of possible square sizes. Theorem 3 gives the lower endpoint explicitly.

For the integer problem, the two constants have different geometric meanings. The constant C is the volume of the three-dimensional region R . The coefficient B comes from the flat boundary pieces $z = 1$, $x = y$, and $y = z$. The remaining two-dimensional boundary pieces are algebraic and curved. Lemmas 6 and 7 show that their total contribution is $o(n^2)$.

We do not know simpler closed forms for C or B . One can also ask the same question for rectangles, or require the three triangle vertices to lie in the relative interiors of the three square sides.

Use of artificial intelligence

OpenAI ChatGPT (GPT-5.6 Sol) was used during the development of this work to explore the geometry, test conjectures, derive and check formulas and arguments, prepare and check verification code, locate related literature, and assist with English wording and editing. The author is responsible for the mathematical content and the final text.

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